

Rodrigo Coimbra

Birthday: 23 April 2002

Current Position: Research Engineer at IST

✉ rodrigo.coimbra@tecnico.ulisboa.pt

🌐 <https://rodscoimbra.github.io/>

in [rodrigo-coimbra](#)

🐙 [RodsCoimbra](#)



About Me

I have been interested in technology since an early age, always looking for the most relevant advances in that area. This curiosity led me to study Electrical and Computer Engineering at IST, where I worked on various projects and became passionate about programming, robotics, and machine learning.

After my bachelor's degree, I pursued a master's specializing in control systems, artificial intelligence, and robotics, with a secondary focus on computing systems. I completed my master's in 2025, and since then I have been working as a research engineer on wheeled and humanoid robotics, focusing on navigation, manipulation, and the integration of advanced planning and learning methods.

Employment History

- 07/2024–09/2024 ■ **Software Developer, Quidgest** - Completed a summer internship as a software developer, working on a project focused on financial management systems. Developed a REST API to enable asynchronous access to database data, implemented in C# for data manipulation and processing, and used basic SQL with Microsoft SQL Server for queries and updates.
- 2025–present ■ **Research Engineer, Institute for Systems and Robotics (ISR)** - Worked on the SocRob@Home project, focusing on navigation for domestic robotics. Responsibilities included developing autonomous navigation for a humanoid robot and contributing to a manipulation pipeline for the team's mobile manipulator robot.

Education

- 2023–2025 ■ **M.Sc. in Electrical and Computer Engineering, IST** - Specialized in control, robotics, and artificial intelligence, with a secondary focus on computing systems. My thesis, on full-body motion planning and control of a mobile manipulator using reinforcement learning, was graded 18. During these two years, I joined the SocRob@Home team, which allowed me to deepen my knowledge and practical experience in robotics. I received an academic merit certificate in each year and completed the master's degree with a final grade of 18.
- 2020–2023 ■ **B.Sc. in Electrical and Computer Engineering, IST** - Completed the degree with a final grade of 16, receiving an academic merit certificate in all three years.

Research Publications

Presentations

- 2026 ■ **ICARSC 2026 Research Presentation** - Presented a paper on full-body local planning using deep reinforcement learning at the ICARSC 2026 conference in Barcelos, Portugal.

Projects

- 2026 – present ■ **euROBIN¹** - euROBIN is a European project whose main goal is to build a network of excellence in AI-powered robotics by bringing together partners from leading universities and research centers, such as the German Aerospace Center (DLR), Karlsruhe Institute of Technology, ETH Zurich, the University of Seville, and Sorbonne University. Within this project, I developed a demonstration shown to the European Parliament, coordinating with other teams to integrate parts of their research into our robot.
- 2025 – present ■ **FOMO-HODOR²** - FOMO-HODOR is a collaboration between the University of Texas at Austin, ISR (Institute for Systems and Robotics), and INESC-ID, focused on researching foundation models for humanoid domestic robots. The project explores the use of large language models in different roles to improve robotics, targeting general-purpose deployment scenarios and exploiting their reasoning capabilities for planning complex tasks. My work centered on developing core capabilities for the robot, including environment mapping and navigation, as well as basic manipulation. I also collaborated closely with researchers from Texas towards shared project goals.
- 2025 ■ **Pilot Project - Oracle & SocRob@Home³** - Project aimed to enhance task and motion planning for domestic robots by integrating Oracle Cloud Infrastructure (OCI).
- 2024 – present ■ **SocRob@Home⁴** - SocRob@Home is an ISR project that researches mobile robots for home environments to assist people with special needs. The team includes Ph.D. students, M.Sc. students, and research engineers who carry out research in these settings and apply the results in international robotics competitions. As part of this project, I was primarily responsible for developing the navigation system of our humanoid robot, as well as the manipulation pipeline for our mobile manipulator.

Events

- 2026 ■ **euROBIN demonstration for the European Parliament – Brussels** - Took part in a cooperative demonstration with other university robotics teams from the euROBIN project, showcasing domestic robotic tasks to members of the European Parliament as an overview of the current state of European robotics.
- **RoboCup@Home – Incheon, South Korea** - Participated as a team member of SocRob in the RoboCup@Home competition, an international event that brings together hundreds of researchers and teams working on service and domestic robotics.
- **ICARSC 2026 - Barcelos** - Attended the International Conference on Autonomous Robot Systems and Competitions, where a paper on full-body navigation was presented.

Outreach

- 2024–2026 ■ **Robot demonstrations for schools** - As part of ISR's science-communication activities, I took part in organized visits to our facilities, presenting our research and running live robot demonstrations for children, teenagers, and the general public.
- 2025–2026 ■ **Project dissemination** - During my time on SocRob, I helped promote the project to students through events such as the JEEC job fair and Técnico Days, explaining the goals of our work.
- 2025 ■ **European Researchers Night⁵** - At Pavilhão do Conhecimento, we showcased our humanoid robot to curious people who wanted to learn more about robotics.

¹<https://www.eurobin-project.eu>

²<https://irsgroup.isr.tecnico.ulisboa.pt/fomo-hodor/>

³<https://blogs.oracle.com/ai-and-datascience/ai-driven-task-motion-planning-service-robots-with-oci>

⁴<https://irs-group.github.io/socrobwebsite/>

⁵<https://www.pavconhecimento.pt/nei/2025>

Outreach (continued)

- **Anniversary of "Pavilhão do Conhecimento"** - During this open-day event, I shared our research and basic concepts in robotics with families and other attendees, using the robot for live demonstrations.

Skills

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| Languages | ■ Strong reading, writing and speaking competencies for Portuguese (native speaker), English (C1). |
| Coding | ■ Python, ROS, ROS2, C++, C, C#, Isaac Sim/LabVIEW, Java, Matlab, SQL. |